4. Implement a finite-state machine for morphological parsing. In this example, we'll create a simple machine to generate plural forms of English nouns using python.

Aim: To implement a simple finite-state machine that generates plural forms of English nouns.

Procedure:

1. Define a noun.
2. Check its ending.
3. Apply the appropriate plural formation rule.
4. Display the singular and plural forms.

Code:

```
def plural(noun):
    if noun.endswith(("s", "sh", "ch", "x", "z")):
        return noun + "es"
    elif noun.endswith("y") and noun[-2] not in "aeiou":
        return noun[:-1] + "ies"
    else:
        return noun + "s"
words = ["cat", "bus", "box", "baby"]
for word in words:
    print(word, "->", plural(word))
```

Output:

```
C:\Users\blohi\Downloads\NLP lab>C:\Users\blohi\AppData\Local\Python\pythoncore-3.14-64\python.exe "c:/Users/b
lohi/Downloads/NLP lab/4.py"
cat -> cats
bus -> buses
box -> boxes
baby -> babies
```

Result: A simple rule-based finite-state model was implemented to generate plural noun forms.